

Note on the Levi-Civita planar regularization of the Three-Body problem

di Riccardo Daluiso

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Let $\Sigma' = Gxy$ be the inertial frame with the origin in the center of mass of the three points. In this system let $\underline{r}_\nu = (x_\nu, y_\nu)$ be the position vectors of the relative distances subjected to the constraint

$$\sum_{\nu=1}^3 x_\nu = 0, \quad \sum_{\nu=1}^3 y_\nu = 0. \quad (1)$$

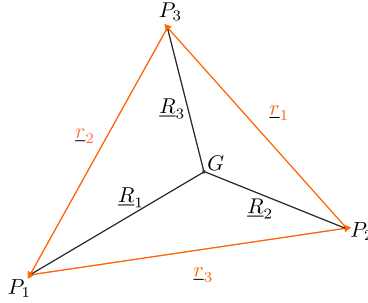


Figure 1: *Mutual distances of the three body in the plane. The origin is in the center of mass G .*

We implement the transformation for $\nu = 1, 2, 3$ defined by

$$x_\nu + iy_\nu = (\xi_\nu + i\eta_\nu)^2, \quad (2)$$

or equivalently

$$\begin{cases} x_\nu = \xi_\nu^2 - \eta_\nu^2, \\ y_\nu = 2\xi_\nu\eta_\nu. \end{cases} \quad (3)$$

If $\rho_\nu^2 = \xi_\nu^2 + \eta_\nu^2$, we have

$$r_\nu = \rho_\nu^2$$

and the expression for U in terms of the new variables becomes

$$U = \gamma M \sum_{\nu=1}^3 \frac{m_\nu^*}{\rho_\nu^2}. \quad (4)$$

The relations (1) can be expressed in terms of the new variables, thus taking the form

$$\sum_{\nu=1}^3 (\xi_\nu^2 - \eta_\nu^2) = 0, \quad \sum_{\nu=1}^3 \xi_\nu \eta_\nu = 0. \quad (5)$$

The constraints above define a 4-dimensional manifold embedded in $\mathbb{R}^6$. In order to prove this, we apply the preimage theorem to the function

$$F : \mathbb{R}^6 \rightarrow \mathbb{R}^2, \\ F(\xi, \eta) = \left(\sum_{\nu=1}^3 (\xi_\nu^2 - \eta_\nu^2), \sum_{\nu=1}^3 \xi_\nu \eta_\nu \right). \quad (6)$$

It is sufficient to show that the matrix

$$J = \begin{pmatrix} 2\xi_1 & 2\xi_2 & 2\xi_3 & -2\eta_1 & -2\eta_2 & -2\eta_3 \\ \eta_1 & \eta_2 & \eta_3 & \xi_1 & \xi_2 & \xi_3 \end{pmatrix}$$

has rank 2. Indeed

$$\det(J^T J) = 4(\rho_1^2 + \rho_2^2 + \rho_3^2)$$

is equal to zero if and only if all the variables ξ_ν, η_ν vanish simultaneously.

From now on we will suppose that the collision arises from P_2 and P_3 . Let $r := r_1$ and $\rho := \rho_1$ so that $r, \rho \rightarrow 0$. By the Implicit function theorem we can use as coordinates $\xi_1, \eta_1, \xi_2, \eta_2$, or $\xi_1, \eta_1, \xi_3, \eta_3$. Instead it is not possible to explicit ξ_1, η_1 as a function of $\xi_2, \eta_2, \xi_3, \eta_3$, as the intuition suggests. In this case, the relevant minor should be

$$\begin{vmatrix} 2\xi_1 & -2\eta_1 \\ \eta_1 & \xi_1 \end{vmatrix} = \rho,$$

which is zero because $\rho = \sqrt{r}$.

Consider

$$\varphi(\xi_\nu, \eta_\nu) = (\xi_\nu^2 - \eta_\nu^2, 2\xi_\nu \eta_\nu)$$

the restriction of the inverse of (2) to the ν -th body, the associated momenta transform as

$$(p_{x_\nu}, p_{y_\nu}) = \left(\frac{\partial \varphi}{\partial(\xi_\nu, \eta_\nu)} \right)^{-T} \begin{pmatrix} p_{\xi_\nu} \\ p_{\eta_\nu} \end{pmatrix},$$

where in this case

$$\left(\frac{\partial \varphi}{\partial(\xi_\nu, \eta_\nu)} \right)^{-T} = \frac{1}{2(\xi_\nu^2 + \eta_\nu^2)} \begin{pmatrix} \xi_\nu & -\eta_\nu \\ \eta_\nu & \xi_\nu \end{pmatrix}$$

Therefore, we obtain the following relations between the old and the new momenta:

$$\begin{cases} p_{x_\nu} = \frac{1}{2\rho_\nu^2} (\xi_\nu p_{\xi_\nu} - \eta_\nu p_{\eta_\nu}), \\ p_{y_\nu} = \frac{1}{2\rho_\nu^2} (\eta_\nu p_{\xi_\nu} + \xi_\nu p_{\eta_\nu}), \end{cases} \quad \nu = 1, 2, 3. \quad (7)$$

In vector form this transformation reads

$$\begin{pmatrix} p_{x_\nu} \\ p_{y_\nu} \end{pmatrix} = \frac{1}{2\rho_\nu^2} \begin{pmatrix} \xi_\nu & -\eta_\nu \\ \eta_\nu & \xi_\nu \end{pmatrix} \begin{pmatrix} p_{\xi_\nu} \\ p_{\eta_\nu} \end{pmatrix}, \quad (8)$$

and it is easily inverted as

$$\begin{pmatrix} p_{\xi_\nu} \\ p_{\eta_\nu} \end{pmatrix} = 2 \begin{pmatrix} \xi_\nu & \eta_\nu \\ -\eta_\nu & \xi_\nu \end{pmatrix} \begin{pmatrix} p_{x_\nu} \\ p_{y_\nu} \end{pmatrix}. \quad (9)$$

Let $\hat{\underline{p}}_\nu := (p_{\xi_\nu}, p_{\eta_\nu})$ and $\underline{p}_\nu := (p_{x_\nu}, p_{y_\nu})$. Then,

$$\hat{p}_\nu := p_{\xi_\nu}^2 + p_{\eta_\nu}^2 = 4\rho_\nu^2(p_{x_\nu}^2 + p_{y_\nu}^2). \quad (10)$$

We now prove that $\hat{p}_\nu$ remains finite at collision for each $\nu = 1, 2, 3$. For this purpose, it is important to relate the relative momenta to the absolute ones. So let $\underline{v}_\nu = \dot{\underline{r}}_\nu$. The kinetic energy in these coordinate reads

$$T = \frac{1}{2} \sum_{\nu=1}^3 m_\nu^* v_\nu^2, \quad (11)$$

where

$$m_1^* = \frac{m_2 m_3}{M}, \quad m_2^* = \frac{m_1 m_3}{M}, \quad m_3^* = \frac{m_1 m_2}{M}.$$

Then

$$\underline{p}_\nu = \frac{\partial T}{\partial \underline{r}_\nu} = m_\nu^* \dot{\underline{r}}_\nu. \quad (12)$$

For notational convenience, we use q_ν to denote one of the coordinates of the ν -th body in the center of mass system of reference, and $V_\nu = \dot{q}_\nu$. Moreover, let $\underline{P}_\nu$ the absolute momenta of the ν -th mass point, we have

$$\begin{aligned} \underline{p}_1 &= \frac{1}{M} (m_2 \underline{P}_3 - m_3 \underline{P}_2) = \frac{m_2 m_3}{M} (\underline{V}_3 - \underline{V}_2), \\ \underline{p}_2 &= \frac{1}{M} (m_3 \underline{P}_1 - m_1 \underline{P}_3) = \frac{m_1 m_3}{M} (\underline{V}_1 - \underline{V}_3), \\ \underline{p}_3 &= \frac{1}{M} (m_1 \underline{P}_2 - m_2 \underline{P}_1) = \frac{m_1 m_2}{M} (\underline{V}_2 - \underline{V}_1). \end{aligned} \quad (13)$$

From the definition of U ,

$$\lim_{t \rightarrow t^*} rU = \gamma m_2 m_3, \quad (14)$$

therefore

$$U = O\left(\frac{1}{r}\right) \quad \text{as } t \rightarrow t^*.$$

From the energy integral

$$\frac{1}{2} \sum_{\nu=1}^3 m_\nu V_\nu^2 = T = U + h,$$

using(14), and because rh tends to 0 at collision,

$$r \sum_{\nu=1}^3 m_{\nu} V_{\nu}^2 = 2r (U + h) \xrightarrow[t \rightarrow t^*]{} 2\gamma m_2 m_3. \quad (15)$$

In particular rV_{ν}^2 , rq_{ν}^2 , $\sqrt{r}q_{\nu}$ for $\nu = 1, 2, 3$ remain finite at collision between P_2 and P_3 . We want to estimate the asymptotic behaviour of V_2, V_3 at collision. Differentiation the center of mass relation $m_1 \underline{R}_1 + m_2 \underline{R}_2 + m_3 \underline{R}_3 = 0$ gives

$$\begin{aligned} rm_2^2 \dot{q}_2^2 &= (\sqrt{r}m_1 \dot{q}_1 + \sqrt{r}m_3 \dot{q}_3)^2, \\ r(m_2 \dot{q}_2)^2 - r(m_3 \dot{q}_3)^2 &= m_1 \sqrt{r} \{m_1 \sqrt{r} \dot{q}_1^2 + 2m_3 \dot{q}_1 (\sqrt{r} \dot{q}_3)\}, \end{aligned}$$

where in the last equation each term on the left side is bounded, $\sqrt{r} \dot{q}_3$ is bounded and $\dot{q}_1$ is finite since P_1 does not collide. Then, as $t \rightarrow t^*$

$$\begin{aligned} r(m_2 \dot{q}_2)^2 - r(m_3 \dot{q}_3)^2 &\rightarrow 0, \\ r(m_2 V_2)^2 - r(m_3 V_3)^2 &\rightarrow 0. \end{aligned} \quad (16)$$

We multiply (15) by m_2 to obtain

$$rm_2 m_1 V_1^2 + rm_2^2 V_2^2 + rm_2 m_3 V_3^2 \xrightarrow[t \rightarrow t^*]{} 2\gamma m_2^2 m_3.$$

Since $rm_2 m_1 V_1^2$ goes to 0, we have

$$rm_2^2 V_2^2 + rm_2 m_3 V_3^2 \xrightarrow[t \rightarrow t^*]{} 2\gamma m_2^2 m_3,$$

and using relation (16)

$$rV_3^2 (m_3^2 + m_2 m_3) \xrightarrow[t \rightarrow t^*]{} 2\gamma m_2^2 m_3,$$

that is

$$rV_3^2 \xrightarrow[t \rightarrow t^*]{} \gamma \frac{2m_2^2}{m_2 + m_3}. \quad (17)$$

Similarly, we obtain the limit for rV_2^2 as

$$rV_2^2 \xrightarrow[t \rightarrow t^*]{} \gamma \frac{2m_3^2}{m_2 + m_3}, \quad (18)$$

and this determines the asymptotic behaviour of V_2 and V_3 as $t \rightarrow t^*$. In particular, $V_{2,3} = O(1/\sqrt{r})$.

Therefore

$$\sqrt{r} \frac{m_2 m_3}{M} (V_3 - V_2) \rightarrow \frac{m_2 m_3}{M} \sqrt{2\gamma(m_2 + m_3)} \underline{e}.$$

where $\underline{e}$ is a unit vector in the collision direction. Taking norms yields

$$r \left| \frac{m_2 m_3}{M} (V_3 - V_2) \right|^2 \rightarrow \frac{2\gamma m_2^2 m_3^2 (m_2 + m_3)}{M^2}.$$

Then

$$p_1 = O\left(\frac{1}{\sqrt{r}}\right) = O\left(\frac{1}{\rho}\right) \tag{19}$$

and from (10)

$$\hat{p}_1 = 2\rho O\left(\frac{1}{\rho}\right) = O(1) \tag{20}$$

remains finite, as we claimed. The same holds for $\hat{p}_2$ and $\hat{p}_3$.